SALLoRA: Robust Parameter-Efficient Adaptation under Structured Multimodal Corruption

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Research Goal

My research investigates how large multimodal foundation models can be adapted efficiently when training data is affected by structured input corruption such as OCR errors. Rather than correcting noisy inputs directly, I study how parameter-efficient adaptation strategies can selectively update model components while preserving robustness and computational efficiency.

Motivation

While modern document understanding models (e.g., LayoutLM-family architectures) are increasingly deployed on real-world scanned documents, the multimodal input stream is frequently degraded by structured OCR corruption. Classical noisy-label learning assumes the input is trustworthy and only the supervision signal is corrupted. My work addresses the inverse challenge: learning robustly when the multimodal input itself is structurally degraded. Standard Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA lack an explicit mechanism to distinguish trustworthy input tokens from corrupted ones. This research proposes shifting from classical noisy-label learning to a noisy-input PEFT paradigm.

Proposed Method

To address structured input noise, I propose SALLoRA, an architecture that filters out corruption during the fine-tuning process without introducing latency during inference. The framework operates through a two-phase process:

- **Phase 1 (Warm-up):** The foundation backbone, LoRA adapters, and layer scalars remain entirely frozen. Only the classifier head is updated to establish a baseline understanding before any low-rank capacity is introduced.
- **Phase 2 (Masked Adaptation):** The framework applies Sensitivity-Aware Layer-wise LoRA. It utilizes decoupled quality estimators (evaluated outside the computational graph to prevent noise from directly shaping the estimator) and a binary sensitivity mask to dynamically evaluate and scale the reliability of input tokens across transformer layers.

Preliminary Observations

Initial experiments were configured on the FUNSD (Form Understanding in Noisy Scanned Documents) benchmark using a frozen LayoutLMv3-base backbone. Synthetic OCR-style corruption was injected strictly into the training set at a 50% corruption level. Preliminary experiments under this synthetic OCR corruption suggest that the proposed framework is more robust than a standard LoRA baseline under the current experimental configuration. These observations remain exploratory and will be validated through controlled baseline comparisons, multi-seed evaluations, and comprehensive ablation studies.

Current Status & Future Work

The current prototype relies on a simplified quality estimator to validate the training framework. To establish definitive conclusions, my immediate research roadmap includes:

- Conducting multi-seed evaluations and calculating dynamic, per-batch sensitivity thresholds rather than relying on static pre-adaptation scaling.

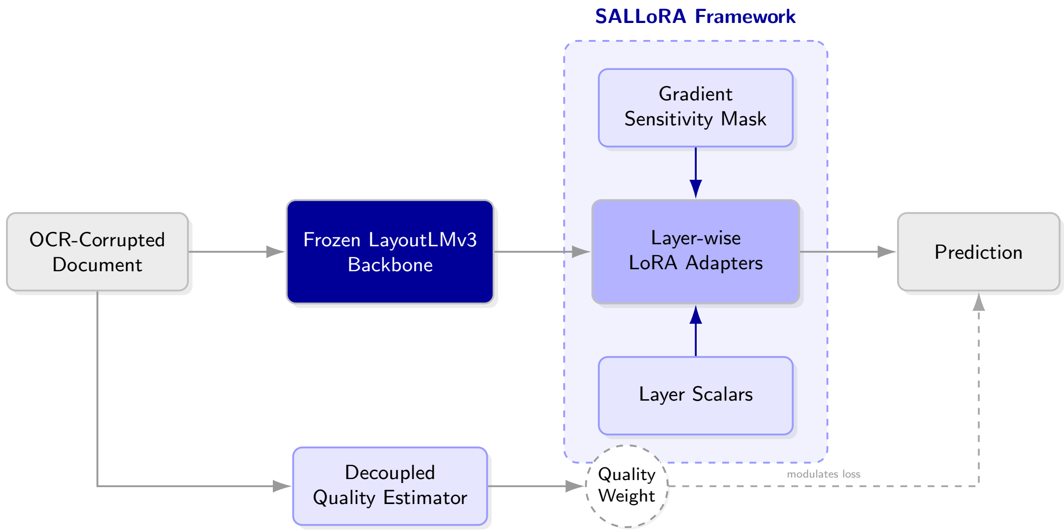

Figure 1: Overview of the proposed SALLoRA framework.

- Integrating a fully supervised or meta-learned quality estimator in place of the current network.
- Running rigorous statistical significance tests and comprehensive ablation studies across varying OCR noise levels against established robust-training baselines.

Long-term Research Vision

Although this work is currently evaluated on document understanding, I envision extending these ideas toward efficient adaptation of large multimodal foundation models, continual fine-tuning, and transferable PEFT methods across language and vision tasks. By decoupling reliable updates from domain-specific noise, I aim to develop robust adaptation strategies that allow base models to continuously learn and transfer knowledge efficiently in noisy, real-world environments.